

KFUPM Business School

Quality Management in Supply Chain Systems

Chapter 14



Managing Quality Improvement Teams & Projects

Master's in Supply Chain Management | KFUPM Business School | Saudi Arabia

Monczka, Handfield, Giunipero & Patterson | Pearson Education

Opening Case: SABIC's Quality Improvement Journey

How a Saudi Giant Transformed Teams into Quality Engines

OPENING CASE

The Challenge

In 2018–2020, SABIC — one of the world's largest petrochemical manufacturers headquartered in Riyadh — faced a critical quality challenge at its Jubail Industrial City complex. Polymer product non-conformance rates reached 3.2% across six production lines, costing the company over SAR 85 million annually in rework, scrap, and customer penalties.

The Solution: Cross-Functional Quality Teams

SABIC deployed structured quality improvement teams — cross-functional groups of engineers, operators, QC specialists, and supply chain managers — using project charters, PERT scheduling, and force-field analysis to drive improvement.

Result: Non-conformance dropped to 0.8% within 18 months, saving SAR 60M/year.

Source: SABIC Sustainability & Performance Report 2020 (sabic.com) | GPCA Quality Benchmarks 2021

SAR 85M

annual quality cost before teams

75% reduction

in non-conformance in 18 months
via structured quality improvement teams



Opening Discussion

Before we dive in — reflect on this:

"Think of a team you have been part of — in university, at work, or in a Saudi company you know. Was it high-performing or dysfunctional? What made the difference? What role did leadership play, and how was conflict handled?"



Pair-Share — 3 minutes, then one spokesperson per group reports back

Why Teams Are Essential in Quality Management

The Drivers of Team-Based Work in Modern Organisations

CONCEPT

Complexity

Modern supply chains span multiple countries, technologies, and regulations. No single person can master all dimensions. Teams pool knowledge.

Knowledge Work

Routine jobs are automated. Employees increasingly do knowledge work — research, analysis, decision-making. This requires autonomy and team input.

Collaboration

Business has moved from command-and-control to networked collaboration. Quality problems cross departmental boundaries — they need cross-functional solutions.

Continuous Change

Saudi Vision 2030 is reshaping entire industries. Teams adapt faster than rigid hierarchies when Aramco, SABIC or NEOM pivot strategy.

Why Employees Thrive in Teams

5 Core Motivators — Research Evidence

Research with project managers across industries identified five consistent motivators for team participation:

1	Mutuality:	Mutual support between management and team members; personal loyalty and encouragement within the team
2	Recognition:	Personal development, achievement recognition, rewards, incentives, and career advancement through team contribution
3	Belonging:	Cohesive and friendly team relationships; clear communication, information sharing, and shared project goals
4	Bounded Power:	Authority and accountability over project resources; ability to influence decisions; personal growth opportunities
5	Creative Autonomy:	Freedom to use creativity and individual potential during the project; good working conditions and intellectual challenge

Saudi Context: Saudi Aramco's TAQNIA employee innovation programme explicitly uses all 5 motivators to drive quality circle participation across its 70,000+ workforce

Situational Leadership in Quality Teams

Hersey, Blanchard & Johnson — Matching Style to Readiness

CONCEPT + VISUAL

S2 — SELLING

High Task, High Relationship

Explain and persuade. Used when team has some skill but low commitment. Leader coaches and builds confidence.

R2: Moderate readiness

S3 — PARTICIPATING

Low Task, High Relationship

Share and facilitate. Used when team is capable but may lack motivation. Leader empowers and collaborates.

R3: Moderate-high readiness

S1 — TELLING

High Task, Low Relationship

Direct and instruct. Used with new team members or novices. Clear, specific guidance. Leader decides.

R1: Low readiness

S4 — DELEGATING

Low Task, Low Relationship

Hand off responsibility. Used when team is fully capable and committed. Leader monitors from a distance.

R4: High readiness

← LOW FOLLOWER READINESS — — — — HIGH FOLLOWER READINESS →

Belbin's 9 Team Roles

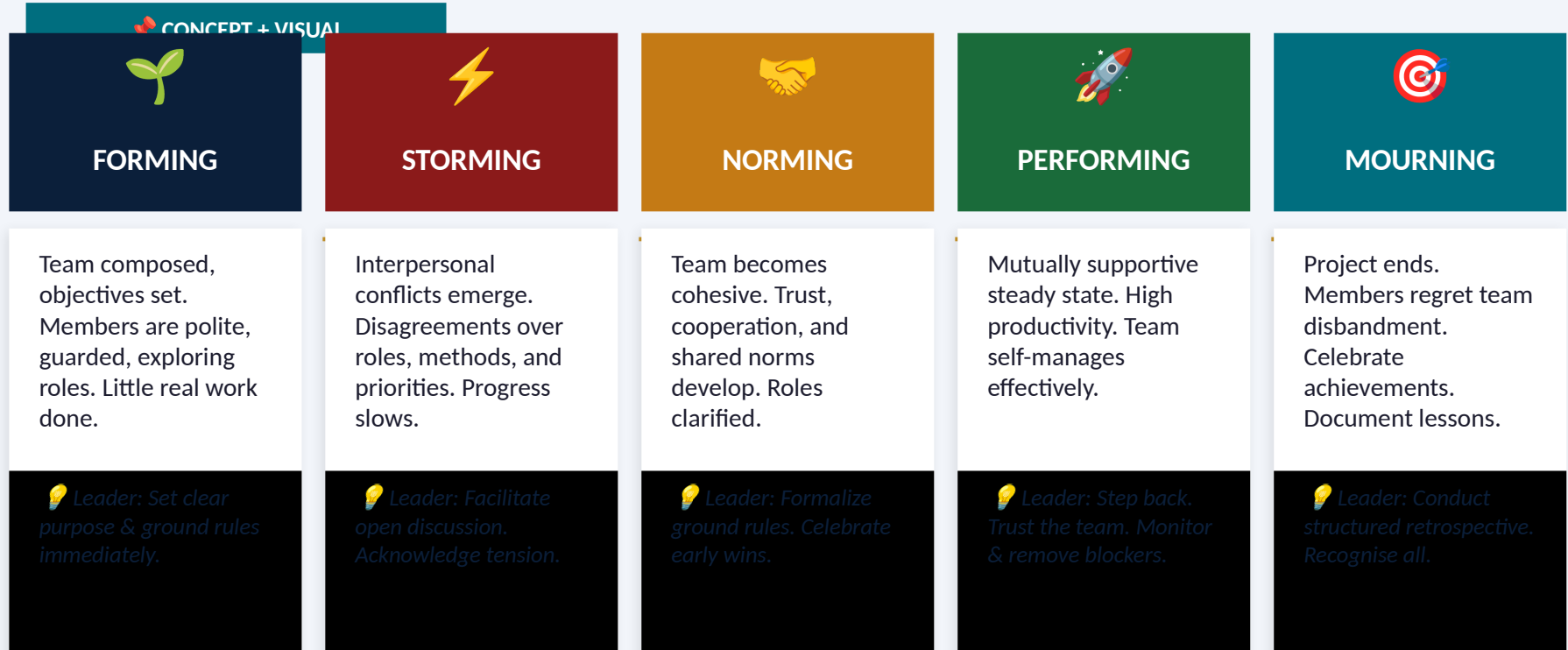
Every High-Performing Team Needs This Balance

Action Roles	Social Roles	Thinking Roles
Shaper Challenges the team, drives action, overcomes inertia	Co-ordinator Delegates, clarifies goals, brings people together	Plant Creative problem-solver, generates novel ideas
Implementer Turns decisions into practical action plans	Team Worker Diplomatic, averts friction, builds team spirit	Monitor-Evaluator Analyses objectively, judges options accurately
Completer-Finisher Ensures no errors, meets deadlines, quality control	Resource Investigator Networks, explores opportunities outside the team	Specialist Deep technical expertise in a specific area

Saudi Application: SABIC project teams audit their composition against Belbin roles at project kickoff to identify gaps (e.g., missing a 'Plant' = low innovation risk).

Stages of Team Development

Forming → Storming → Norming → Performing → Mourning



Aramco Team Experience: A new IKTV-compliance quality team at Aramco reportedly spent 3 months in 'Storming' before structured facilitation accelerated them to 'Performing'.

Types of Quality Improvement Teams

Choosing the Right Team Structure for the Problem

Process Improvement Team

Works to improve a specific process or customer service. May be management-directed or self-directed. Duration: ongoing.

 Saudi Electricity Company (SEC): PIT reduced meter-reading errors by 62% in 2022.

Cross-Functional Team

People from different functional areas (engineering, QC, supply chain, finance). Tackles issues crossing departmental borders.

 SABIC assembled CFTs across procurement, ops, and QC to address polymer packaging defects.

Tiger Team

High-powered, short-duration team assigned to a specific crisis. Intense focus. Used in reengineering or urgent problem-solving.

 NEOM project: Tiger team deployed when structural materials failed inspection — 72-hour resolution mandate.

Self-Directed Work Team

Team charts its own improvement projects. Minimal management oversight. High autonomy and accountability.

 Almarai dairy plant: SDWTs reduced milk spoilage by 45% after being given full process control authority.

Virtual Team | Team members do not meet physically. Use collaborative tools (Teams, SAP, Slack). Spans geographic or organisational boundaries. |  Aramco–Shell joint venture quality teams operate virtually across Saudi Arabia, Netherlands, and US.

Empowerment & Team Ground Rules

Building the Foundation for High Performance



Empowerment — What It Means

Empowerment = giving team members genuine decision-making authority over their own work and processes.

Implicit management promises to employees:

- Greater control over your own work
- No penalty for making painful changes
- Management trusts you and values your ideas
- You will be rewarded for decisions that benefit the company
- Management will concede control over company systems to you

Preconditions for empowerment to work:

- Clear authority and accountability
- Participation in planning at all levels
- Adequate information for decision-making
- Responsibility matched with authority



Ground Rules for Effective Teams

- Attend all meetings or notify the facilitator in advance
- Start and end meetings on time — respect everyone's schedule
- One person speaks at a time — no side conversations
- Critique ideas, never people — maintain mutual respect
- Everyone participates — no passengers on quality teams
- Decisions by consensus wherever possible
- Document all agreements and assigned actions
- Confidentiality within the team on sensitive issues
- No phones or distractions during active sessions

Real Case Study: Saudi Aramco Quality Circle Programme

World-Class Teams Driving Operational Excellence

Background

Saudi Aramco — the world's largest oil producer — operates one of the most sophisticated quality improvement team (QIT) programmes globally. Since the 1990s, Aramco has run formal Quality Circles and Cross-Functional Improvement Teams across its upstream, downstream, and support

Key Team Management Practices:

- Structured project charters signed by business unit champions
- PERT/Gantt scheduling for all QIT projects > 3 months
- Force-field analysis used before every major process change
- Belbin role assessment at team formation
- Formal conflict resolution protocol embedded in team SOPs
- Cost-benefit analysis required for all project proposals > SAR 500K
- Post-project retrospective with documented lessons learned

Source: Saudi Aramco Annual Report 2022 | Aramco Quality & Reliability Management Framework (aramco.com)

Programme Results

400+

active QITs at any given time

SAR 2.8B

cumulative savings (2015–2022)

96%

project charter completion rate

<6 months

average team project duration

87%

of QITs achieve ≥80% of stated goal

35,000+

employees trained in QIT methods

Conflict Resolution in Quality Teams

20%+ of Leader Time — Master This Skill

4 Stages of Conflict: Frustration → Conceptualisation → Interaction → Outcome



Avoidance

Let things work out on their own. Appropriate for minor, low-stakes conflicts. Risk: issues fester and escalate.

Low assertiveness, Low cooperativeness



Defusion

Smooth ruffled feelings while getting the project back on track. Focus on team cohesion over winning.

Low assertiveness, High cooperativeness



Win-Win

Find solutions satisfying both parties. Balance demands: identify what each person wants and look for mutual solutions.

High assertiveness, High cooperativeness



Confrontation

Leader injects themselves; uses active listening and structured problem-solving. Best when defusion fails.

High assertiveness, Low cooperativeness



Choosing a Winner

When differences are irreconcilable, leader chooses a course of action and ensures both parties commit.

High assertiveness



Preventing Conflict

Proactive organizational design: align goals, rewards, communication, and coordination to eliminate conflict sources.

Strategic / systemic

The Project Charter: Quality Team's Contract

Defining Objectives, Scope, and Accountability

A Project Charter is the foundational document that officially authorises a quality improvement project. It aligns team, management, and stakeholders.

Problem Statement

Clearly state what is wrong — use data.
E.g., 'Outbound delivery error rate is 4.2% vs. industry benchmark of 0.8%, costing SAR 3.2M/year in rework and ...'

Scope / Boundaries

Define what is IN and OUT of scope.
E.g., 'Covers Riyadh DC outbound operations only. Excludes supplier inbound quality.'

Expected Benefits

Quantified financial + quality benefits.
Hard savings: SAR X/year. Soft savings: customer satisfaction improvement.

Timeline & Milestones

Kickoff: Week 1 | Define complete: Week 2 | Measure complete: Week 5 | Analyze: Week 8 | Improve pilot: Week 12 | Control: Week 18

Project Objective

Measurable goal with deadline. E.g., 'Reduce delivery error rate to $\leq 1.0\%$ within 6 months, saving $\geq$ SAR 2.5M/year.'

Primary Metric (CTQ)

The one number that defines success.
E.g., 'Monthly delivery error rate (%) measured at customer receipt confirmation.'

Team Members & Roles

Black Belt (leader), 2 Green Belts, Operations Supervisor, QC Technician, IT Lead. Champion: DC Director.

Qualifying Projects: Cost-Benefit Analysis

Should We Do This Quality Project? Here's How to Decide

CBA Formulas

Total Project Cost:

$$C_t = \sum (C_d + C_i)$$

C_d = direct costs, C_i = indirect costs

Payback Period:

$$PP = C_t \div B_a$$

Saudi Example: SABIC WMS Project

Direct Costs: SAR 420,000

(Software: 180K | Hardware: 90K | Consulting: 150K)

Indirect Costs: SAR 185,000

(Staff time 120K | Training 65K)

Total Cost: SAR 605,000

✓ Solution

Total Cost = SAR 605,000 | Annual Benefits = SAR 2,400,000 | Payback Period = $605,000 \div 2,400,000 = 0.25$ years (≈ 3 months) | ROI = 297%

⚠ Hard Costs vs. Soft Costs

Hard costs = real SAR savings (reduced labour, materials, penalties).
Soft costs = efficiency gains without direct SAR savings (e.g., faster noncritical tasks). Justify projects on HARD costs only.

📌 Rule of Thumb

Any quality project with $PP < 1$ year and $ROI > 150\%$ is considered high priority at Aramco and SABIC project selection gates.

Force-Field Analysis: Will the Project Succeed?

Identify Driving & Restraining Forces Before You Start

TOOL + EXAMPLE
Force-Field Analysis: List all forces FOR and AGAINST a change. Score each 1 (weak) to 5 (very strong). Sum both sides. If restraining forces dominate — address them first.

Example: Implementing a Digital Quality Management System at Almarai Dairy, Riyadh (2021)

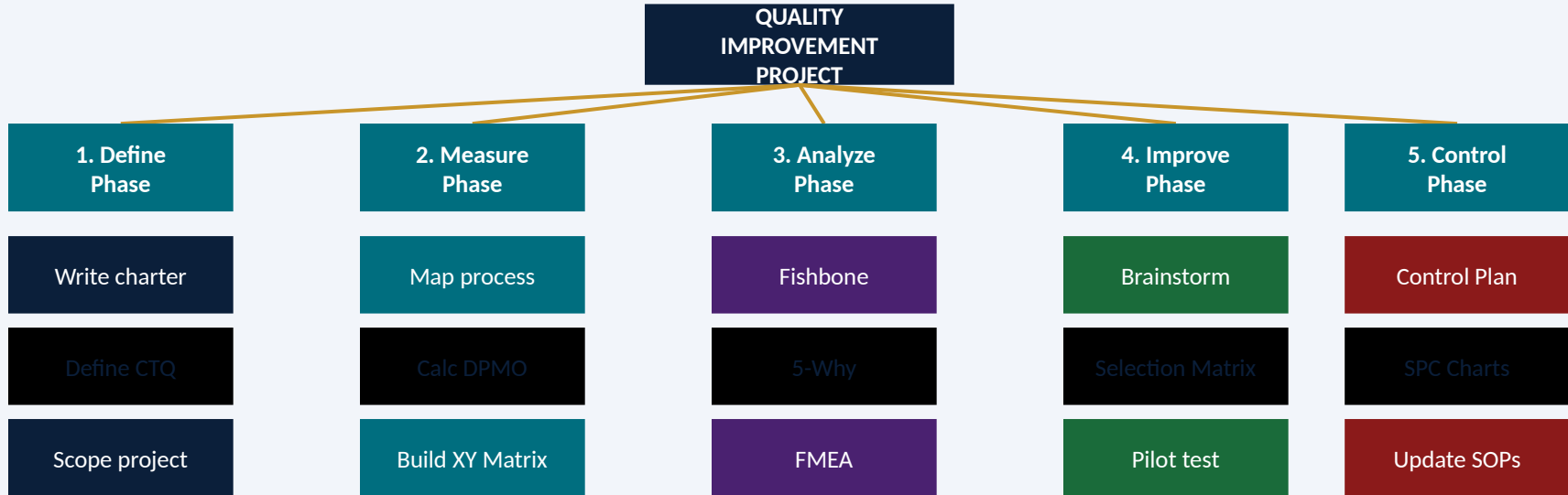
✓ DRIVING FORCES (Total: 22)		✗ RESTRAINING FORCES (Total: 18)	
Senior management mandate from CEO	5	Resistance to digital change	4
SAR 8M annual rework savings projected	4	Cost and time: SAR 1.2M	4
Saudi FDA compliance requirement (2022 deadline)	4	Senior managers oppose system	5
IT infrastructure already in place	4	Integration with legacy ERP complex	3
Successful pilot at Al-Kharj plant	3	Older data format incompatibility	2

Driving: 22 vs. Restraining: 18 | Verdict: DRIVING forces dominate ✓ — Project can proceed. Address the 2 senior managers and ERP integration first.

Work Breakdown Structure (WBS)

Decomposing the Quality Project into Manageable Tasks

A WBS decomposes a project into its component parts. Each task is placed on a note card, precedence relationships identified, and the network drawn.



Task Time Formula: $t = (a + 4m + b) \div 6$ where a=optimistic, m=most likely, b=pessimistic time estimates

Total project time T = sum of critical path task times. Slack = Late Start - Early Start. Tasks with zero slack form the critical path.

PERT Network & Critical Path Analysis

Scheduling the Quality Project — Saudi DC Quality Example



NUMERICAL EXAMPLE

Scenario: Quality Improvement project at a Saudi pharmaceutical distribution centre (Riyadh). 7 tasks identified.

Task	Description	Pred.	a	m	b	$t=(a+4m+b)/6$
A	Write project charter	—	2	3	8	3.33
B	Map current process	A	3	5	10	5.50
C	Collect defect data (30 days)	A	28	30	35	30.50
D	Analyse root causes (fishbone+5-why)	B,C	5	7	14	7.50
E	Develop solutions + selection matrix	D	4	6	10	6.33
F	Pilot test (2-week trial)	E	10	14	21	14.17
G	Implement control plan + SOPs	F	5	7	12	7.33

Critical Path: A → C → D → E → F → G | Total = 3.33 + 30.50 + 7.50 + 6.33 + 14.17 + 7.33 = 69.16 days (≈ 14 weeks) | Task B has slack of 25 days



Group Discussion Activity

Building Quality Teams in Saudi Industry

Scenario:

You are the Quality Director at Ma'aden (Saudi Arabian Mining Company) in Ras Al-Khair Industrial City. Your smelting plant has experienced a 15% increase in aluminium sheet defects over the past 6 months, costing SAR 22M in scrap and rework. You need to form and lead a quality improvement team to investigate and fix this problem within 9 months.

Your Tasks (Groups of 3-4, 15 minutes):

1. What TYPE of team should you form? (Process Improvement / Cross-Functional / Tiger Team / etc.) Justify your choice.
2. Using Belbin's model, identify FOUR roles you must have on this team and what each person would contribute.
3. List the GROUND RULES you would establish at the first meeting. Which 3 are most critical in a Saudi workplace context?
4. The VP of Operations refuses to release two key engineers for the team. Which CONFLICT RESOLUTION mode do you apply, and what specifically do you say?





In-Class Simulation



The NEOM Quality Team Challenge — Project Charter & PERT Scheduling

Context

NEOM's Smart City project requires a quality management system for its 120 km TRON supply chain corridor. As an engineering procurement and construction (EPC) project, the quality of incoming construction materials — steel, cement, electrical components — is critical. Defect rates in the first 6 months of procurement have reached 6.8% versus a target of 1.0%.



TEAM A: Project Team

- Form a cross-functional quality team (name 5 roles)
- Write a complete Project Charter (all 7 elements)
- Calculate the payback period for a SAR 780K quality system that saves SAR 3.2M/year
- Perform force-field analysis (identify 4 forces FOR and 4 AGAINST)



TEAM B: Scheduling Team

- Break the project into 6 tasks with predecessor relationships
- Assign PERT times (optimistic/most likely/pessimistic) to each task
- Calculate task times using $t = (a + 4m + b) \div 6$
- Identify the critical path and total project duration



Role cards distributed now. 20 minutes to prepare. Then each team presents their deliverable — 4 minutes each.

TEAM A ROLE CARD — Project Charter & CBA Team — CONFIDENTIAL

NEOM Supply Chain Quality Project | You are the Project Management Team

Task 1 **Team Formation**

Name 5 team roles using Belbin's model. Specify each person's background (engineer, QC specialist, supply chain manager, etc.) and what they contribute to the project.

Task 2 **Project Charter**

Complete all 7 charter elements: Problem Statement | Objective (measurable) | CTQ Metric | Scope | Expected Benefits | Team Roles | Timeline (6 milestones).
Data: Defect rate 6.8% → target 1.0%. Monthly procurement value SAR 85M. Each defect costs ~SAR 45,000 in rework/rejection.

Task 3 **CBA Calculation**

Digital QMS implementation cost: Software SAR 420K, Consulting SAR 180K, Training SAR 80K, System integration SAR 100K. Annual benefit: SAR 3.2M (defect reduction) + SAR 850K (process efficiency).
Calculate: Total cost, Annual benefit, Payback Period, ROI.

Task 4 **Force-Field Analysis**

Identify 4 driving forces (FOR the project) and 4 restraining forces (AGAINST). Score each 1–5. Sum both totals. Is the project ready to proceed, or must restraining forces be reduced first?

● TEAM B ROLE CARD — PERT Scheduling Team — CONFIDENTIAL

NEOM Supply Chain Quality Project | You are the Project Scheduling Team

Task	Pred.	Description	a (opt)	m (likely)	b (pess)	t = (a+4m+b)/6
A	—	Write project charter & team setup	2	3	8	Calculate:
B	A	Process mapping + data collection	20	25	35	Calculate:
C	A	Supplier quality audit (5 key vendors)	10	14	21	Calculate:
D	B,C	Root cause analysis (fishbone + FMEA)	5	8	14	Calculate:
E	D	Develop & pilot solutions	15	20	28	Calculate:
F	E	Implement control plan + SOPs	7	10	16	Calculate:

Instructions:

1. Calculate t for each task using the formula $t = (a + 4m + b) \div 6$
2. Draw the network (A → B+C → D → E → F) and identify the critical path
3. Calculate total project duration (sum of critical path task times)
4. Identify which task has the most slack (Non-critical path)

 **DO NOT SHARE THIS CARD WITH TEAM A — Present your PERT network at the 20-minute mark**

Simulation Debrief: NEOM Quality Team Challenge

Model Answers & Key Learning Points

✓ Team A Model Answer — Charter

Problem: Defect rate 6.8% vs. target 1.0% → SAR 22.1M/year COPQ
Objective: Reduce to $\leq 1.0\%$ within 9 months
CTQ: Outbound defect rate (%) measured at NEOM receiving gate
Payback: $(420K + 180K + 80K + 100K) \div (3,200K + 850K) = \text{SAR } 780K \div 4,050K = 0.19 \text{ years } (\approx 70 \text{ days})$
ROI = $(4,050K - 780K) \div 780K \times 100 = 419\%$

✓ Team B Model Answer — PERT

A: 3.33d

B:
25.83d

C:
14.17d

D: 8.17d

E:
20.50d

F: 10.50d

Critical Path: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow F = 3.33 + 25.83 + 8.17 + 20.50 + 10.50 = 68.33 \text{ days}$

Task C has slack: $(3.33 + 25.83) - (3.33 + 14.17) = 11.66 \text{ days}$

Total project: $\approx 14 \text{ weeks}$

Key Teaching Points from the Simulation:

- The charter is not bureaucracy — it is the team's contract. Without it, scope creep destroys projects.
- PERT critical path reveals that data collection (Task B, 25.83 days) controls the whole timeline — a common real-world trap.
- ROI of 419% in 70 days is a no-brainer — yet many Saudi organisations delay such projects due to poor project qualification processes.
- Force-field analysis prevents 'project assassins' — identify them in planning, not after they kill the project.

Real Case: Ma'aden's Quality Project Management System

Mining Giant Deploys Structured Quality Teams at Scale

Background

Saudi Arabian Mining Company (Ma'aden) operates aluminium smelting, phosphate, and gold mining across Saudi Arabia. As the company expanded its Ras Al-Khair Industrial City operations (2018–2022), it faced systemic quality challenges: equipment failure rates, supplier non-conformance, and process deviations were threatening production targets.

Quality Team Management Practices Deployed

- Project charters required for ALL quality projects > SAR 200K
 - Cross-functional teams mandatory for issues crossing 2+ functions
 - PERT scheduling tool integrated into Ma'aden SAP PM module
 - Force-field analysis: standard pre-implementation gate review
 - Conflict resolution: formal 3-step escalation protocol
 - Belbin role assessment at team formation (HR-facilitated)
 - Cost-benefit analysis: minimum ROI threshold of 200%
- Source: Ma'aden Annual Report 2022 (maaden.com.sa) | Ma'aden Quality & Safety Framework 2021

Outcomes (2020–2022)

SAR 1.4B	savings from quality improvements
42%	reduction in equipment failure rate
67%	reduction in supplier non-conformance
180	quality improvement projects completed
97%	of projects met schedule within 10%
4.6/5	employee team satisfaction score

- Tiger teams: deployed for safety-critical issues within 48 hrs

Virtual Quality Teams & Digital Collaboration

Managing Distributed Teams in Saudi's Vision 2030 Era

Saudi Vision 2030 is creating geographically distributed supply chains — NEOM, Red Sea Project, Qiddiya. Quality teams increasingly operate across Riyadh, Jubail, Yanbu, Jeddah, and international partners simultaneously.

What Changes in Virtual Teams

- Status differences & social hierarchies diminish — junior members contribute more equally
- Normal visual and auditory cues are lost — tone, body language, silence become ambiguous
- E-negotiators make more aggressive initial positions — anchor high/low more readily
- Real-time vs. asynchronous creates coordination challenges across time zones
- Risk of 'social loafing' increases — accountability must be made explicit

Best Practices for Virtual Quality Teams

- Establish explicit virtual ground rules in the first session (camera on, response time SLAs)
- Use SAP, Microsoft Teams, or Slack with dedicated project channels per quality initiative
- Daily 15-min stand-ups replace long weekly meetings for active improvement phases
- Assign a 'virtual facilitator' whose only role is managing participation and energy
- Physical kickoff (once) then virtual — trust built face-to-face sustains virtual collaboration

Managing Multiple Quality Projects Simultaneously

Coordination, Prioritisation & Resource Allocation

Quality in Supply Chain CONCEPT International quality teams are rarely managing just one project. Aramco, SABIC, and NEOM simultaneously run 50–400 quality improvement projects.

Project Name	Champion	Black Belt	Status	Progress
Project A: Supplier Defect Reduction	VP Supply Chain	Ahmed Al-Ghamdi	Month 3/9	65%
Project B: Warehouse Error Rate	DC Director	Sara Al-Otaibi	Month 1/6	20%
Project C: Delivery Time Improvement	Logistics VP	Khalid Al-Rashidi	Month 5/6	85%
Project D: Customer Returns Analysis	Quality Director	Nora Al-Dosari	Planning	10%

⚠ Management duty: Ensure no employee serves on more than 2 active quality teams simultaneously — cognitive overload reduces effectiveness. Track this on the multiple-project control form.

Saving Quality Teams from Failure

Diagnose & Intervene Before It's Too Late

Warning Signs: Cost overruns, missed deadlines, finger-pointing, team members 'bailing out', frustration and blame escalating — diagnose EARLY, intervene DECISIVELY.

✗ Unclear project scope

Symptom: Team argues about 'whose job this is' constantly

✓ Return to charter. Reconfirm scope and boundaries with champion present.

✗ Conflict unresolved

Symptom: Two members refuse to work together; meetings become combative

✓ Leader applies confrontation mode: one-on-one sessions, then structured mediation.

✗ Management disengaged

Symptom: Champion stops attending reviews; decisions waiting for weeks

✓ Escalate immediately. Present financial impact of delay to business unit head.

✗ Wrong people on the team

Symptom: Key decisions stall because expertise is missing

✓ Belbin assessment reveals gap. Recruit missing role. Reassign non-contributors.

✗ No data collected

Symptom: Analysis phase stalls; everything is 'we think' not 'we know'

✓ Revert to MEASURE phase. Define data plan, assign ownership, set 2-week deadline.

✗ Resistance to change

Symptom: Solutions agreed in meetings are ignored when implemented

✓ Use force-field analysis to identify resisters. Involve them in solution design.

Quality Team Management: Decision Guide

Which Tool for Which Situation?

SYNTHESIS

 You need to select the most impactful quality project	Cost-Benefit Analysis + Payback Period
 Stakeholders are resisting the quality initiative	Force-Field Analysis → Strengthen driving forces
 Team is stuck in 'Storming' and making no progress	Situational Leadership S2 (Selling) + Ground Rules reset
 Team lacks a critical functional expertise (e.g., no IT person)	Belbin Role Assessment → Identify gap → Recruit
 Two team members are in direct conflict	Confrontation mode → Active listening → Win-Win solution
 Project is running over schedule	PERT Network → Identify critical path → Crash or rescope
 You need to formally document and authorise the project	Project Charter (all 7 elements + champion signature)
 Multiple quality projects competing for same staff	Multiple-Project Control Form → Limit staff to ≤ 2 teams

Quality Teams & Saudi Vision 2030

Giga-Projects Demand World-Class Team Management

Saudi Vision 2030's mega-projects (NEOM, Red Sea Project, Qiddiya, ROSHN) involve the most complex supply chains and quality requirements in Saudi history. Every tool in this chapter is being deployed at scale right now.

NEOM / Line Project

Challenge: 1,500 km of linear city construction requires cross-functional quality teams across 200+ contractors. PERT scheduling is mandatory for all Tier-1 EPC contracts.
 *Project charters + PERT networks signed at contract award.*

Saudi Aramco IKTV

Challenge: IKTV mandate requires domestic suppliers to meet world-class quality standards. Aramco runs formal quality team training programmes for all qualifying vendors.
 *Vendor quality circles + force-field analysis for IKTV compliance projects.*

SABIC / PetroRabigh

Challenge: Joint venture quality improvement teams span Saudi, Japanese (Sumitomo), and US team members — virtual multi-cultural quality teams are now standard.
 *Virtual team protocols + cross-cultural conflict resolution frameworks.*

Ma'aden Expansion

Challenge: Phase 3 Ras Al-Khair expansion: tiger teams deployed when construction quality incidents occur. Formal diagnosis-intervention cycle embedded in project contracts.
 *Tiger teams + diagnosis-intervention cycle + CBA for all quality projects > SAR 500K.*

Exit Ticket — Before You Leave

Consolidate Your Learning

EXIT ACTION

1. Name the 5 stages of team development

...and describe what a LEADER should do at each stage.

2. What is a Project Charter?

...list 4 of its 7 elements and explain why each matters.

3. Calculate the payback period

...if a quality project costs SAR 1.2M and saves SAR 4.8M/year.

4. A team member is consistently late and non-contributing.

...which conflict resolution mode do you apply first? Why?

5. Pose ONE question you still have

...post to the LMS discussion board before midnight.



Write answers on a notecard and hand in, OR post to LMS before midnight tonight.

Key Takeaways

- 1 Teams outperform individuals on complex quality problems — especially cross-functional challenges spanning procurement, ops, and QC.
- 2 Situational leadership is NOT one-size-fits-all. Match your style (Telling/Selling/Participating/Delegating) to your team's readiness.
- 3 Teams progress through 5 predictable stages. Storming is normal — your job as leader is to facilitate it, not avoid it.
- 4 The Project Charter is the team's contract. Without it, projects fail from scope creep, unclear goals, and resource conflicts.
- 5 Cost-benefit analysis must use hard costs. A payback period < 1 year and ROI $> 150\%$ = priority quality project.
- 6 Force-field analysis reveals project killers BEFORE they strike. Identify restraining forces in planning, not during execution.
- 7 PERT networks expose the critical path — the sequence of tasks that determines how long the project actually takes.
- 8 In Saudi Vision 2030 giga-projects, structured quality team management is not optional — it is contractually mandated.